

Problem

Determine a two-port network that is represented by the following z parameters:

$$[z] = \begin{bmatrix} 6 + j3 & 5 - j2 \\ 5 - j2 & 8 - j \end{bmatrix} \Omega$$

Step-by-step solution

Step 1 of 2

$$(Z) = \begin{bmatrix} 6 + j3 & 5 - j2 \\ 5 - j2 & 8 - j \end{bmatrix} \Omega$$

Since $Z_{12} = Z_{21}$ This two port network is symmetric

$$\begin{aligned} Z_{11} - Z_{12} &= 6 + j3 - 5 + j2 \\ &= 1 + j5 \end{aligned}$$

$$\begin{aligned} Z_{22} - Z_{12} &= 8 - j - 5 + j2 \\ &= 3 + j \end{aligned}$$

Step 2 of 2

